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MYP Science Command Terms – Made Simple

All 40 command terms, explained in plain English with a science example

How to use this: The command term tells you **what to do** and **how much to write**. The groups go from the shortest answers (top) to the deepest thinking (bottom). Underline the command term in every question before you start.

⚡ Short answers

Just give the answer. No explanation needed.

Term	What it means	Example question	Example answer
📖 **Define**	Give the exact meaning of a word or idea.	Define speed.	Speed is the distance travelled per unit of time.
🔎 **Identify**	Pick the right answer, or name one feature.	Identify the force that pulls objects towards Earth.	Gravity.
📝 **List**	Write several short answers, with no explanation.	List three renewable energy sources.	Solar, wind, hydroelectric.
📏 **Measure**	Find the value of something using an instrument.	Measure the length of the pencil.	14.5 cm (using a ruler).
🧠 **Recall**	Remember something you learned before.	Recall the unit of force.	The newton (N).
☑️ **Select**	Choose from a list or group.	Select the conductor: wood, copper, plastic.	Copper.
💬 **State**	Give a short answer: a name, a value or a fact.	State what happens to ice when it is heated.	It melts.
✍️ **Write down**	Give the answer, usually taken straight from the information. Little or no working.	Write down the temperature at 5 minutes from the table.	42 °C.

🔢 Calculations

Use numbers. Show your working.

Term	What it means	Example question	Example answer
🧮 **Calculate**	Work out a number answer and show every step.	Calculate the speed of a car that travels 100 m in 5 s.	v = d ÷ t = 100 m ÷ 5 s = 20 m/s.
🎯 **Determine**	Find the one possible answer.	Determine the resistance if V = 12 V and I = 3 A.	R = V ÷ I = 12 ÷ 3 = 4 Ω.
≈ **Estimate**	Find an approximate value.	Estimate the height of the classroom.	About 3 m (about twice my height).

🔍 **Find**	Get an answer and show the working that matters.	Find the mass of 2 m^3 of water (density 1000 kg/m^3).	m = ρ × V = 1000 × 2 = 2000 kg.
👣 **Show**	Write out every step of a calculation or proof.	Show that 36 km/h equals 10 m/s.	36 km/h = 36 000 m ÷ 3600 s = 10 m/s.
🔓 **Solve**	Find the answer using the correct method.	Solve for t: 50 = 10t.	t = 50 ÷ 10 = 5.
✅ **Verify**	Give evidence that proves an answer is correct.	Verify that 4 Ω is correct for V = 12 V, I = 3 A.	V = I × R = 3 × 4 = 12 V, which matches.

✏️ Drawing and graphs

Show it as a picture, diagram or graph.

Term	What it means	Example question	Example answer
🗒️ **Annotate**	Add short notes to a diagram or graph.	Annotate the graph to show where the car stops.	Arrow at the flat part: 'speed = 0, car has stopped'.
🏗️ **Construct**	Show information as a diagram or in a clear structure.	Construct a circuit diagram with a battery, switch and bulb.	Draw the symbols joined in one loop.
📐 **Draw**	Make an accurate diagram or graph: pencil, ruler, labels and scale.	Draw a graph of distance against time.	Labelled axes with units, plotted points, line of best fit.
🏷️ **Label**	Add names or titles to parts of a diagram or graph.	Label the parts of the eye.	Lens, retina, pupil, iris, cornea.
📍 **Plot**	Mark points accurately on a graph.	Plot the results from the table.	A small cross for each pair of values.
〰️ **Sketch**	Draw a quick labelled graph or diagram showing the general shape.	Sketch how temperature changes as ice melts.	Line rises, goes flat at 0 °C, then rises again.

🗂️ Organizing information

Sort, arrange or give the main points.

Term	What it means	Example question	Example answer
🧺 **Classify**	Sort things into groups.	Classify these as solid, liquid or gas: ice, steam, water.	Solid: ice. Liquid: water. Gas: steam.
📚 **Document**	Give credit to your sources in the text and in a reference list.	Document the websites you used.	In-text citations plus a reference list in one style.
🗄️ **Organize**	Put ideas or data in a clear order.	Organize your results.	A table with headings and units.

📊 **Present**	Show information so others can see it clearly.	Present your data.	A titled table and a labelled graph.
📌 **Summarize**	Give only the main idea or main points.	Summarize the experiment.	Heavier objects did not fall faster than lighter ones.

💡 Describing and explaining

Write more. Give details and reasons.

Term	What it means	Example question	Example answer
🗒️ **Outline**	Give a short summary of the main points.	Outline how a thermometer works.	Liquid expands when heated and rises up the tube.
🖼️ **Describe**	Give a detailed picture of what happens.	Describe the motion of a ball dropped from a height.	It starts at rest, speeds up as it falls, and hits the ground at its fastest.
💡 **Explain**	Say what happens and WHY. Give reasons or causes.	Explain why metal feels colder than wood.	Metal is a good conductor, so it takes heat from your hand faster.
🧩 **Interpret**	Use what you know to find patterns and draw conclusions from data.	Interpret the graph of current against voltage.	Straight line through zero, so current is proportional to voltage.
🧪 **Demonstrate**	Prove something using reasons, evidence or an example.	Demonstrate that air has mass.	An inflated balloon weighs more than an empty one.
🧱 **Formulate**	State an idea or argument clearly and precisely.	Formulate a hypothesis about pendulum length.	If the string is longer, then the swing takes longer, because...

⚖️ Thinking and judging

Think deeply. Use evidence. Make a decision.

Term	What it means	Example question	Example answer
🔬 **Analyse**	Break information into parts to find what it shows.	Analyse the results of the ramp experiment.	Speed rose with angle, but trial 3 is an outlier.
🔧 **Apply**	Use what you know in a new situation.	Apply Newton's 3rd law to a rocket launch.	Gas pushes down, so the rocket is pushed up.
🗨️ **Comment**	Give a judgement on a statement or result.	Comment on the result g = 12 m/s².	Too high (expected 9.8 m/s²); probably a timing error.
🛠️ **Design**	Plan an experiment, model or solution.	Design an experiment to test how height affects bounce.	Variables, equipment, method, safety, data table.

🗣️ **Discuss**	Look at both sides, use evidence, then give a conclusion.	Discuss using nuclear power.	Low CO_2 but produces waste; overall useful if waste is stored safely.
⚖️ **Evaluate**	Weigh strengths and weaknesses to make a judgement.	Evaluate your method.	Repeats made it reliable; reaction time was a weakness. Use light gates.
🛡️ **Justify**	Give valid reasons or evidence to support your answer.	Justify your choice of a digital thermometer.	More precise (0.1 °C) and easier to read than a liquid one.
🤔 **Suggest**	Propose a possible answer, idea or improvement.	Suggest one improvement to the experiment.	Use a light gate instead of a stopwatch.

Based on the IB MYP command terms for sciences. Definitions simplified for students.